

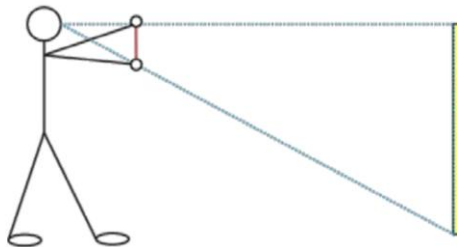
PHYS002A Lab 1 - Measurements and Uncertainties

Objectives

- To make measurements with uncertainty and down to the correct number of significant digits
- To understand how to reduce uncertainties
- To use percent difference in the comparison of two measurements.

Overview

Using the **method of similar triangles**, one is able to determine the dimensions of an object based on the dimensions of known values. For example, one is able to determine the length of a lawn by looking at a piece of string (red in the figure) and an object of a known size placed at the opposite end of the lawn (yellow in the figure) because the ratio of the distance between the eye and stick and the distance between the eye and string must be equal to the ratio of the length of the stick and the length of the string.



$$\frac{\text{Distance from eye to stick}}{\text{Distance from eye to string}} = \frac{\text{size of stick}}{\text{size of string}}$$

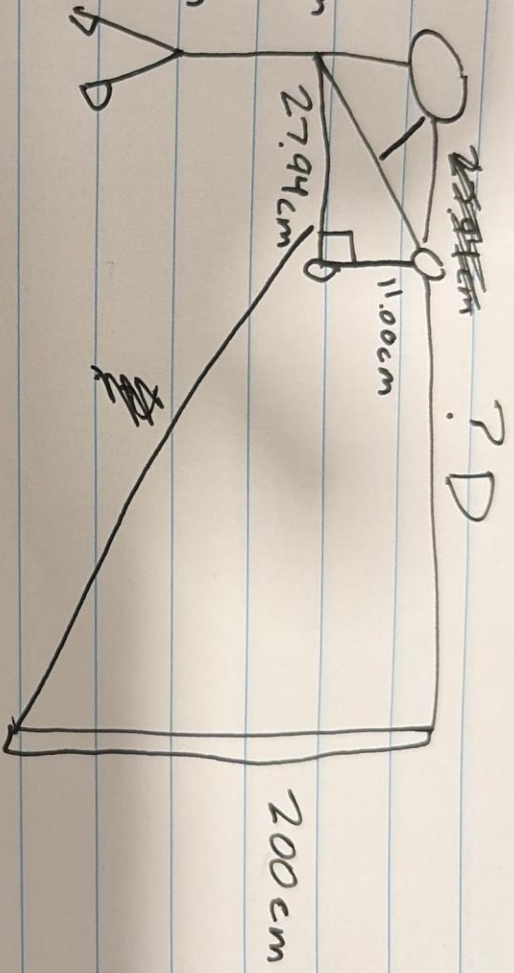
[Sketch, Label and Insert screenshot]

Measuring the size of an object through this **indirect method** can be useful for large and/or distant objects, however this method is much less precise and with greater uncertainty than a direct measurement.

You will apply this indirect method using the appropriate uncertainties, and COMPARE the results with a **direct measurement**.

$$\frac{D}{27.94} = \frac{200}{11 \text{ cm}}$$

$$D = 508 \text{ cm}$$



Background on measurements and uncertainty

In Module 0 in canvas, find the “Basic data analysis-2” page. Read the sections “Measurement Uncertainties” and “Significant figures” for information on determining the uncertainty of a measurement and reporting measurements to the correct number of significant figures.

Equipment (list the equipment for your group)

Procedure

You will need three people in your lab group to complete this activity, partner A, partner B, and partner C. If you have 4 people in your group, rotate activities so that everyone contributes.

Indirect Measurement

- Go out to the lawn near the lab room.
- Partner A holds a 2-meter stick vertically on one edge of the lawn. (If a 2-meter stick isn't available, then combine two 1-meter sticks to make one).
- While standing on the opposite edge of the lawn, partner B holds a string at arm's length from their eye. Using the string and their fingers, partner B measures how long the 2-meter stick appears to be from their position.

1. Record the apparent size of the stick with appropriate units and precision.

The apparent size of the stick is 11 inches (27.94 cm) from the distance across the lawn.

2. What is the uncertainty of this first measurement? Explain.

The uncertainty of the first measurement is approximately $\pm 9/16$ of an inch

I found that the finger that was used to calculate the measurement was $9/16$ of an inch wide, so this means that I have to guess to make my measurement any closer than within $9/16$ in.

3. Why may the precision and uncertainty be different? Explain.

Precision is the smallest increment that is available on the measuring instrument, while uncertainty is a representation of how accurately the measurement tool or method

works. I can write that the stick appeared to be 11 inches (27.94 cm) from the distance across the lawn. However, this method gives approximately 9/16 inches (1.43 cm) of uncertainty.

- Partner A (I don't have a partner C) measures the distance between the eye of partner B and the string when partner B measures the apparent size of the stick.
- 4. Record the distance between the eye and the string with precision and uncertainty**
19 15/16 +/- 1/32 inch (774.70 cm +/- 0.08)
The uncertainty was 1/2 inch (1.27 cm)

5. Explain how your group determined the units, precision and uncertainty.

My precision came from the tape measure we were using to measure the units that contained 1/16 inch ticks as the smallest unit of measurement.

The uncertainty came from the thickness of my finger that was used to measure the tape measure during the different steps (different fingers were used to simulate different group members). Since our fingers were 9/16 inches (1.43 cm) and 1/2 inches (1.27 cm) thick, this means that we can not accurately locate the exact measurement on the tape measure any closer than 9/16 inch (1.43 cm) and 1/2 inch (1.27 cm) respectively.

The units are being converted into centimeters because of the fact that we are measuring a two-meter stick. This way, all the measurements are easily comparable.

6. Using the previous measurements, determine the length of the lawn by calculating the distance between partner A's 2-meter stick and partner B's eyes.

Based on the previous measurements, the length of the lawn is 5.55×10^3 cm long. (I hope this is supposed to be off)

I did the calculations on paper $\rightarrow (27.94)(200/11) = 508$ cm.

7. Name several sources of uncertainties and rank them in size and quantify them. Calculate the percent uncertainty using Equation 1.7 in the textbook. Show your work. Hint: "Basic Data Analysis" in Module 0 may be useful.

1. The biggest uncertainty was when calculating the apparent stick size on the string. The uncertainty of +/- 1.43 cm due to my finger thickness caused marking errors when measuring.

2. The eye-to-string distance was also a strong point of uncertainty, as it is hard to get an exact measurement from someone's eye to the exact end of their arm where they would hold the string. Also, once again, the finger thickness plays a factor in this, as the group member has to mark the measurement. Using a smaller finger makes the uncertainty smaller, around +/- 1.27 cm. The precision for both of these measurements is +/- 1/32 in (0.08 cm).
3. The last uncertainty is that the two-meter stick may not have been held straight up and down. I could predict that this leads to an uncertainty of +/- 0.50 cm.

Calculations with Percent Uncertainty Equation (I think)

$$(0.5/200)+(1.0/11.0)+(1.27/27.94) = 0.139 = 13.9\% \text{ uncertainty}$$

8. What is the uncertainty on the size of the lawn (the answer to question 7)?

$$(0.5/27.94)+(0.5/200)+(1.27/11.0) = 0.139 = \pm 13.5\% \text{ uncertainty}$$

$$13.5\% \times 508\text{cm} = \pm 68.6 \text{ cm} = \pm 70\text{cm approximately}$$

9. Do you think the actual size of the lawn falls within your range of estimated sizes? Explain.

I know that the actual size of the lawn does not fall within the range of my estimated sizes because of the method of measuring the stick and string, as well as the fact that I'm not sure if I did it properly.

My thought process is that my answer should, in fact, be within the range of my estimate because the uncertainty helps create the range based on the flaws in the experiment.

10. What factor most influenced your estimate of the size of the lawn? Explain.

The factor that most influenced my estimate of the size of my lawn was the uncertainty in my initial measurements and the thickness of my fingers while marking the string to measure the 2-meter stick.

Direct Measurement

- Use the tape measure to measure the distance between partner A's 2-meter stick and the spot on the ground directly below partner B's eyes.

11. What is the direct size of the lawn with uncertainty?

The direct measurement is 538.48 cm with uncertainty of +/- 0.08cm

12. How did you determine the uncertainty in the direct measurement?

Since my measurement instrument has the smallest unit of measurement at $\pm 1/16$ inch, you take half of that per side and say $\pm 1/32$ in = ± 0.08 cm

13. Which measurement do you think is more reliable, the measurement made with the string or the measurement made with the measuring tape? Why?

The measurement made with the tape measure is more reliable because it relies on fewer calculations, which leaves less room for human/rounding error. Also, the tape measure used has a smaller uncertainty than when my finger is marking the piece of string.

14. When comparing each measurement, is the measurement of length using the string consistent with the measurement of length using the measuring tape? That is, does one value agree with the other if each of their experimental uncertainties is taken into account?

They should stay consistent in their measurements as long as the experimental uncertainties are taken into account and the calculations are done without error. The indirect value of 508 cm is within the ± 68.6 cm of the direct measurement of 538cm.

15. What is the percent difference between the length of the lawn measured using the string and the length of the lawn measured using the measuring tape?

The percent difference between the length when measured using the string vs the measuring tape is 5.7%

$$(538\text{cm} + 508\text{ cm}) / 2 = 523$$

$$538 - 508 = 30$$

$$30/523 = 0.057 = 5.7\%$$

16. If you were to repeat the experiment, will you get the same length for the lawn?

Once again, the results will be consistent with the uncertainty. However, the exact measurements may vary on the indirect measurements due to human error and rounding error.

Average, Spread, Reported Average

Simulate this by going to each group and recording the length of the lawn that each group measured. (Direct and Indirect). **For the sake of this assignment, ChatGPT will generate 5 other group measurements.**

17. Record the length of the lawn measured by each group (label by group #).

Group 1 (you): Direct 538.48 cm, Indirect 5.08 m

Group 2: Direct 534.67cm, Indirect 5.30 m

Group 3: Direct 541.02 cm, Indirect 5.55 m

Group 4: Direct 536.88 cm, Indirect 5.00 m

Group 5: Direct 543.56 cm, Indirect 5.62 m

Group 6: Direct 532.77 cm, Indirect 4.95 m

18. Calculate the average length of the lawn measured by each group.

$(538.48 + 534.67 + 541.02 + 536.88 + 543.56 + 532.77) / 6 = 537.90$

19. Calculate the spread (Max - Min).

$543.56 - 532.77 = 10.79 \text{ cm}$

20. Divide the spread by 2 to calculate the standard error on the average.

$10.79 / 2 = \text{Standard Error} = 5.40 \text{ cm}$

21. Report the best estimate as average +/- standard error, down to the correct number of significant figures.

The best estimate is $537.90 \pm 5.4 \text{ cm}$

22. How close is your length measured with the string to the best estimate?

$$537.9 - 508.00 = 29.9\text{cm}$$

My measured length with the string was 29.90 cm below the best estimate.

23. If you were to redo this experiment, how could you minimize your uncertainty?

If I redo this experiment, I could use a thin marker to mark the string rather than my finger to reduce the uncertainty in that aspect. Also, I can use a tape measure with mm markings rather than 1/16 inch markers on it. I had to use that due to the time constraint to complete the assignment and the lack of materials at my disposal. It would also be easier to measure my lawn and do all the measurements during daylight (this was completed after sundown). I could also repeat the indirect measurement for multiple trials and take the average to get my indirect measurement closer to the direct measurement.

Note: I hope I did this lab properly. The next one will be better, and I will understand the criteria and material better. Thank you for your understanding.